MontGoWork

AI-Powered Workforce Navigator for Montgomery, Alabama

From "I can't get a job" to a personalized, barrier-aware re-entry plan — with real Montgomery resources, transit schedules, benefits screening, and AI-generated action steps a resident can use **Monday morning**.

1,900+	40	30	7
TESTS PASSING	PRS MERGED	SPRINTS SHIPPED	BARRIER TYPES

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The Problem

Montgomery residents re-entering the workforce face interconnected barriers that generic job boards ignore. A single barrier — like no car or a criminal record — can disqualify someone from 60% of available jobs. Multiple barriers compound exponentially.

Fragmented Resources

Career centers, childcare providers, credit counselors, and legal aid exist — but there's no single system that connects them to a resident's specific situation.

Benefits Cliffs

Taking a job at the wrong wage can cost more in lost benefits than the paycheck provides. A $1/hr raise can mean losing $400/month in childcare subsidy.

Transit Blindness

Jobs that look great on paper are unreachable without a car. No Sunday bus service. Last bus at 9 PM kills second-shift jobs. Generic job boards don't know this.

Criminal Record Barriers

Alabama residents with records don't know which employers are fair-chance, whether they qualify for expungement, or how to navigate background check timing.

7 Barriers

MontGoWork addresses credit, transportation, childcare, housing, health, training, and criminal record barriers in a single integrated system

How MontGoWork Works

1

Assess

5-minute wizard captures ZIP, barriers, credit profile, criminal record, household income, and work history.

→

2

Match

Engine scores jobs by Practical Value Score (net income + proximity + schedule + barrier fit). Screens 7 benefits programs. Checks transit feasibility.

→

3

Plan

AI generates a personalized "Monday Morning" plan with barrier cards, ranked jobs, benefits eligibility, cliff warnings, and exportable PDFs.

Montgomery-Specific

Real M-Transit routes (14 routes, 42 stops), local career centers, childcare providers, and training programs — not national aggregations.

No Account Required

No sign-up, no PII stored permanently. Session-based with 30-day expiry. Credit data stays in browser only. Privacy by design.

Case Worker Ready

Career Center Ready Package generates a staff-facing PDF with WIOA eligibility, document checklist, and "what to say" scripts.

Maria's Journey

Maria, 34, recently released. Single mother of 2 in Montgomery (36104). No car. Credit score ~580. Criminal record (misdemeanor theft, 4 years ago). Four barriers.

Assessment Wizard (3 minutes)

ZIP 36104, unemployed, no vehicle. Selects credit, transportation, childcare, and criminal record barriers. Self-reports credit at 580, 65% utilization. Misdemeanor theft, 4 years ago, sentence complete. Work history: retail + food service, forklift certified.

Matching Engine Runs

PVS scores jobs (4-factor weighted). Credit filter splits eligible-now vs. after-repair. Transit schedule matcher validates shift times against M-Transit first/last bus. Benefits screener checks SNAP, childcare subsidy, Medicaid, Section 8. Cliff calculator detects a $200 drop at $14/hr. Expungement check: eligible under Alabama Act 2021-507.

"Monday Morning" Plan Generated

AI writes a personalized narrative with Montgomery-specific guidance. Four barrier cards with timelines and local resources. Ranked job matches with fair-chance badges and transit routes. Benefits eligibility dashboard. Cliff chart showing safe wage ranges. Expungement filing steps.

Export & Take Action

Downloads PDF to bring to the Alabama Career Center on Ripley Street. Career Center Ready Package gives the case worker a staff-facing summary, WIOA eligibility, and a "what to say" script. QR code on the PDF links to a 30-day follow-up feedback form.

Key Features

Practical Value Score (PVS)

Jobs ranked by real-world value: 35% net income (with benefits cliff impact), 25% proximity, 20% schedule fit, 20% barrier compatibility. No-pay jobs penalized with 0.55x multiplier. Fair-chance employers boosted.

SCORING ENGINE

Benefits Cliff Detection

Calculates net income across $8-$25/hr in $0.50 steps for 7 Alabama programs. Flags severe cliffs (>$200 loss) so residents and case workers avoid wage traps. Visual cliff chart in the plan.

FINANCIAL SAFETY

Transit Schedule Matching

14 M-Transit routes, 42 stops. Validates shift times against first/last bus. Detects transfers via Rosa Parks hub. Flags no-Sunday and no-night service. Walk distance to nearest stop. Google Maps trip planning link.

MONTGOMERY-SPECIFIC

Criminal Record Intelligence

Alabama Act 2021-507 expungement eligibility check. Fair-chance employer matching. Background check timing awareness (pre-offer vs. post-offer). Excluded charge category filtering.

LEGAL GUIDANCE

AI-Powered Plan Narrative

Multi-provider LLM (Claude, OpenAI, Gemini) generates Montgomery-specific "Monday Morning" action plans. Template fallback if APIs are down. Prompt injection defense. Rate limiting.

AI / LLM

Resource Ecosystem

Local career centers, childcare providers, training programs, social services — with findhelp.org deep links, eligibility badges, feedback-driven health scoring, and barrier graph traversal for root-cause analysis.

COMMUNITY

Architecture

FRONTEND

- **Next.js 15** (App Router)
- React + TypeScript
- Tailwind CSS + shadcn/ui
- TanStack React Query
- html2pdf.js (PDF export)
- WCAG 2.1 AA accessible
- Mobile-responsive

BACKEND

- **FastAPI** + Uvicorn
- Python 3.11+
- SQLAlchemy (async) + aiosqlite
- SQLite database
- FAISS + sentence-transformers (RAG)
- Multi-provider LLM client
- Matching engine + PVS scorer
- Benefits cliff calculator
- Transit schedule matcher
- Barrier graph + causal traversal

EXTERNAL SERVICES

- Anthropic Claude API
- OpenAI API
- Google Gemini API
- BrightData (job scraping)
- JSearch / RapidAPI
- Honest Jobs (fair-chance)
- Credit Microservice
- findhelp.org (resource links)
- EmailJS (plan delivery)

1,499	469	100%	14
Backend Tests	Frontend Tests	TDD via PairCoder	DB Tables

Graceful degradation: app works without any external service. Template narratives, seed jobs, and local resources provide full functionality offline.

What Sets Us Apart

vs. Generic Job Boards

Capability	Job Boards	MontGoWork
Transit feasibility	None	**Schedule-aware**
Benefits cliff warnings	None	**7 programs**
Criminal record matching	Checkbox filter	**Policy-level matching**
Credit barrier awareness	None	**Filter + repair path**
Barrier-aware scoring	Keyword match	**4-factor PVS**
Local resource connection	None	**Barrier cards + findhelp**
Case worker handoff	None	**Career Center PDF**

Design Principles

✓ **Privacy First:** No accounts, no PII storage. Session-based with 30-day expiry. Credit data in browser sessionStorage only.

✓ **Graceful Degradation:** Every external service has a fallback. AI down? Template narrative. No BrightData? Seed jobs. No credit API? Skip credit scoring.

✓ **City-Portable:** Architecture is city-agnostic. Swap seed data (resources, transit, employers) and prompts for any city. Zero code changes.

✓ **Feedback Loop:** Resource health scoring via user feedback. Watch/Flagged/Hidden statuses. 30-day decay detection. QR-linked post-visit surveys.

✓ **Security Hardened:** Rate limiting, CORS, prompt injection defense, admin auth, input validation, Docker hardening, CI vulnerability scanning.

Built With & What's Next

Completed (30 Sprints)

Core Platform
Assessment wizard, matching engine, plan generation, PDF/email export

AI Integration
Multi-provider LLM narratives, barrier intelligence RAG chat, prompt guardrails

Job Intelligence
BrightData scraping, PVS scoring, fair-chance matching, credit filters

Benefits Engine
7-program eligibility screening, cliff calculator, visualization

Transit System
Schedule-aware matching, walk distance, transfer detection, Sunday/night gaps

Security & Quality
1,900+ tests enforced by PairCoder TDD gates, security audit, Docker hardening, accessibility

Roadmap

findhelp.org API Integration
Deep API partnership for real-time resource availability and eligibility

Multi-City Expansion
Deploy for Birmingham, Mobile, Huntsville — swap seed data, same codebase

PostgreSQL + Redis
Production database, caching layer, horizontal scaling for concurrent users

Case Worker Dashboard
Admin portal for career center staff to track outcomes and manage resources

Employer Portal
Fair-chance employers can register policies, receive matched candidate referrals

Outcome Tracking
30/60/90-day employment outcomes via feedback loop. Measure real impact.

Built with PairCoder — the enforcement layer that made it possible

PairCoder is an AI pair programming framework that enforced TDD (tests before code), architecture constraints (400-line file limits, 50-line function limits), acceptance criteria gates, and role-based workflows (Navigator/Driver/Reviewer) across every sprint. It managed task lifecycle with Trello sync, budget validation, and context persistence across sessions — ensuring 30 sprints, 40 PRs, and 1,900+ tests shipped with consistent quality.

PairCoder · Python · FastAPI · Next.js · React · TypeScript · SQLite · Claude API · FAISS · Docker